

Face-Vertex Problem

Def'n: For a polytope P , define the
function: $FV: \{\text{faces}\} \times \{\text{vertices}\} \rightarrow \{0,1\}$

$$FV(f,v) = \begin{cases} 1 & \text{if } v \in f \\ 0 & \text{if } v \notin f \end{cases}$$

Alice: face f

Bob: vertex v

Extended Formulations and Comm C.

Lemma: If P has an extended formulation Q with r inequalities, then the non-deterministic communication complexity of $\text{FACE-VERTEX}(P)$ is at most $\log_2 r$.

Notation: $Q = \{(x, y) : Cx + Dy \leq d\}$
 $\pi(Q) = P$ denote the affine map.

Proof — 1/2

Stack matrix encodes the matrix of FV

Specifically: $\boxed{\text{supp}(S) \text{ encodes FV.}}$

Recall: $S_{ij} = b_i - a_i^T x_j$

By Yannakis's result, ~~the~~ ^{an} extended formulation with r inequalities gives a non-neg. factorization:

$$S = \begin{bmatrix} u \\ \vdots \end{bmatrix}_f \begin{bmatrix} v^T \\ \vdots \end{bmatrix}_r$$

We show that non-det. comm. compl $\leq \log_2 r$

~~$S_{f,v}$~~ $S_{f,v} = \underline{U_f} \cdot V_v$ by non-negativity of U, V

$$U_f \cdot V_v > 0 \quad \underline{\text{iff}} \quad \exists j \in \{1, \dots, r\} \text{ s.t.}$$

$$U_{f,j}, V_{r,j} > 0$$

Proof — 2/2

This gives a comm protocol:

- Prover sees $f \in V$ and announces index $j \in \{1, \dots, r\}$
- Alice accepts iff $U_{fj} > 0$
- Bob accepts iff $V_{vj} > 0$.



Hence: We can lower bound the complexity of an extended formulation by lower bounding the non-det comm complexity of FACE-VERTEX.

Another Proof — 1/2

Suppose we have $FV(f, v) = 1$, i.e., Bob's vertex v is not in Alice's face f .

|| If prover can announce an index — in this case a facet G of the extended formulation Q (s. $G \in \{1, 2, \dots, r\}$) st:

$$(i) \exists y_v \text{ s.t. } (v, y_v) \notin G \quad (\pi^{-1}(v) \cap Q \not\subseteq G)$$

$(v, y_v) \in Q \text{ and}$

$$(ii) \forall (x, y) \in Q \text{ with } x \in f, (x, y) \in G \quad (\pi^{-1}(f) \cap Q \subseteq G)$$

then facet G of Q proves $v \notin f$, i.e., $FV(f, v) = 1$.

⊗ Both Bob & Alice can check (i) & (ii).

We have left to show that if $v \notin f$, indeed such a facet G of Q exists.

Another Proof — 2/2

$$\begin{aligned}\pi^{-1}(f) \cap Q &= \{(x, y) \in Q : x \in f\} \\ \pi^{-1}(v) \cap Q &= \{(v, y) \in Q\}\end{aligned} \quad |$$

$$\text{Since } v \notin f \Rightarrow (\pi^{-1}(f) \cap Q) \cap (\pi^{-1}(v) \cap Q) = \emptyset$$

FACT: f, v faces of $P \Rightarrow \pi^{-1}(f) \cap Q, \pi^{-1}(v) \cap Q$
are faces of Q . // Exercise.

FACT: Any face of a polytope is the intersection of faces that define it.

$$\text{Thus: } \pi^{-1}(f) \cap Q = G_{i_1} \cap G_{i_2} \cap \dots \cap G_{i_k}, \quad i_\ell \in \{1, \dots, r\}$$

Since $\pi^{-1}(v) \cap Q$ does not intersect $\pi^{-1}(f) \cap Q$, $\pi^{-1}(v) \cap Q$ cannot be contained in each G_{i_ℓ} , so there is at least one, call it G
s.t. $\pi^{-1}(v) \cap Q \not\subseteq G$ (property (i))

However, $\pi^{-1}(f) \cap Q = G_{i_1} \cap \dots \cap G_{i_k} \Rightarrow \pi^{-1}(f) \cap Q \subseteq G$ (property (ii)).